

Definitions:

$V(t)$ = average daily volume across all stocks per day in corresponding market cap category.

$\sigma(t)$ = average stock volatility in the month.

SPX = SP500 index value on last day in month.

$\Delta V(t)$ = log change in daily volume (MOM).

$$\Delta V(t) = \ln\{V(t)\} - \ln\{V(t-1)\}$$

$\Delta V(t-1)$ = previous month's log change in daily volume (MOM) to incorporate an autoregressive term. This is also a proxy for momentum.

$$\Delta V(t-1) = \ln\{V(t-1)\} - \ln\{V(t-2)\}$$

$\Delta V(t-12)$ = log change in daily volume (MOM) one year ago to incorporate a monthly pattern.

$$\Delta V(t-12) = \ln\{V(t-12)\} - \ln\{V(t-13)\}$$

$\Delta\sigma_{large}(t)$ = log change in large cap volatility (MOM).

$$\Delta\sigma_{large}(t) = \ln\{\sigma_{large}(t)\} - \ln\{\sigma_{large}(t-1)\}$$

$\Delta\sigma_{small}(t)$ = log change in small cap volatility (MOM).

$$\Delta\sigma_{small}(t) = \ln\{\sigma_{small}(t)\} - \ln\{\sigma_{small}(t-1)\}$$

$\Delta Spx(t)$ = log change in SP500 index value (MOM). We used the change in SP500 index values for both large cap and small cap forecasts.

$$\Delta Spx(t) = \ln\{spx(t)\} - \ln\{spx(t-1)\}$$

Monthly Volume Forecasting Model:

$$\Delta V(t) = b_0 + b_1 \cdot \Delta V(t-1) + b_2 \cdot \Delta V(t-12) + b_3 \cdot \Delta\sigma + b_4 \cdot \Delta Spx$$

Figure 7.5 shows the average daily volume per stock in each month for large cap and small cap stocks over the period Jan. 2002 through Dec. 2011 (10 years of monthly data). For example, in Jun. 2011 the average daily volume for a large cap stock was 5,766,850 per day and the average daily volume for a small cap stock was 555,530 per day. It is important to note that historical volume levels will change based on stock splits and corporate actions.